

The Elements of DeFi

DeFi does not have an innovation problem. It has a vocabulary problem.

Nearly every protocol ever shipped is a recombination of about fifty recurring mechanisms — and once you can name them, most of the industry stops looking novel and starts looking legible.

Call it a periodic table if you like, but it is not one: no periodic law, no atomic number, nothing recurring down a column. Chemistry discovered its elements. These were designed, in a hurry, by people shipping to production — which is exactly why the combinations are the part worth writing down.

THEME

Dark Light

v1.0 · reviewed 2026-08-04

48 core · 10 candidate · 10 contested · 29 laws · 19 hazards

READ A PROTOCOL

Pick one. It lights up in the table below, numbered in the order each part became necessary. Four of these are dead — and those are the instructive ones.

Aave v3 Pooled lending 10 elements	Uniswap v3 Spot exchange 4 elements	Maker / Sky CDP stablecoin 10 elements	Liquity v1 CDP stablecoin 6 elements
GMX v2 Perpetual futures 8 elements	Lido v2 Liquid staking 5 elements	CoW Protocol Intent exchange 4 elements	CCTP v2 Fast Cross-domain transfer 4 elements
Centrifuge Real-world credit 8 elements	Terra / Anchor + Algorithmic stablecoin 6 elements	Mango Markets + Perpetual futures 6 elements	Euler v1 + Pooled lending 7 elements

Rows are how much must already exist

S0 needs nothing but the ledger. S4 needs other people to agree. Deeper is not more dangerous — it is more dependent, and confusing those two is how you misprice a protocol.

Colour only ever means hazard

There is exactly one coloured thing in this interface and it is danger. Everything else earns its distinction from position, shape and lightness.

Blocks are families of substitutes

Everything in a block answers one question. Swap Cp for Cl and you still have a DEX. That is what a block means and the only thing it means.

The shape is the finding

Families cluster at one depth — credit is all S3, pricing all S1. And S3 is enormous: most of DeFi is machinery for owing, pricing and unwinding obligations.

VIEW

The table

Laws

Hazards

ARRANGEMENT

By depth

By family

DIMENSION

Flat

3D

READING ORDER

By group

By stratum

By ID

SEARCH

symbol, name or ID

Row = prerequisite depth, exactly. **Blocks** inside a row are families of substitutes. Order within a block is **not** semantic.

S0 Ledger-local claims 3 elements

G01 Claims & accounting3

E001S0ShPro-rata share accountingcore

E002S0IxIndex-based accrualcore

E003S0RbRebasing accountingcore

S1 Deterministic transformation 10 elements

G02 Pool pricing6

CSMS1Constant sumn limit

E004S1CpConstant-product...core

E005S1WgWeighted-geometric...core

E006S1StStable-hybrid invariantcore

E007S1CLConcentrated liquiditycore

E008S1PmOracle-priced inventory curvecore

G03 Execution2

E009S1ObOn-chain order bookcore

E010S1RfRequest for quotecore

G04 Liquidity catalysis2

E039S1AgAggregation & routingcore

E040S1FLAtomic flash liquiditycore

S2 Measured or time-conditioned 14 elements

G03 Execution1

E011S2

Ba

Batch-auction clearing

core

G08 Truth4

E028S2

Ex

External data oracle

core

E029S2

Tp

Time-weighted price

core

E030S2

Oa

Optimistic assertion...

core

E031S2

At

Reserve / NAV attestation

core

G09 Time & queueing3

E032S2

Sr

Streaming accrual

core

E033S2

Ep

Epoch-gated transition

core

E034S2

Wq

Withdrawal queue

core

G10 Incentives2

E035S2

Em

Protocol-funded emissions

core

E058S2

Fd

Surplus & fee distribution

candidate

G14 Access & privacy4

E046S2

Aw

Permission / identity gate

core

E047S2

Sb

Shielded-balance state

core

E048S2

Sd

Selective-disclosure...

candidate

E056S2

Fz

Freeze / forced transfer

candidate

S3 obligation & solvency 22 elements

G05 Credit5

E013S3

Pl

Pooled lending

core

E014S3

Im

Isolated lending market

core

E015S3

Cd

Collateralized-debt minting

core

E016S3

Uc

Undercollateral credit

core

E017S3

Ft

Fixed-term debt

core

G06 Solvency5

E018S3

Ct

Collateral-threshold test

core

E019S3

Li

Incentivized liquidation

core

E020S3

Ad

Auto-deleveraging

core

E021S3

Sl

Socialized-loss allocation

core

E022S3

Bs

Staked backstop

core

G07 Risk transfer7

E023S3

Pf

Perpetual funding...

core

E024S3

Op

Option payoff

core

E025S3

Tr

Tranche waterfall

core

E026S3

Cv

Mutual cover pool

core

E027S3

Py

Principal/yield separation

core

E057S3

Sv

Servicing & determination...

candidate

E060S3

Dp

Directional position &...

candidate

G11 Control & authority1

G13 Stability3

G16 Staking1

E051S3

Gs

Sponsored-fee liability

n

candidate

E043S3

Rd

Direct redemption...

core

E044S3

Ps

Peg-swap module

core

E045S3

As

Algorithmic supply...

core

E059S3

Vl

Staking & validator...

n

candidate

S4Multi-agent & cross-domain10 elements

G03Execution1

E012S4

In

Intent & solver execution

core

G11Control & authority4

E036S4

Tg

Delayed-governance...

core

E037S4

Up

Mutable implementation...

core

E038S4

Gp

Guardian or pause

core

E050S4

Au

Delegated execution scope

core

G12Cross-domain4

E041S4

Xm

Cross-domain message...

core

E042S4

Xf

Cross-domain asset transfer

core

E053S4

Rl

Resource lock / reservation

n

candidate

E054S4

Of

Optimistic fill & reimbursement

n

candidate

G15Security reuse1

E049S4

Rs

Restaking / shared security

n

candidate

Contested register – 10 entries

Held outside the table because a necessary criterion is genuinely disputed.
Not usable in a formula without a note. The separation means “argued about”, not “filed away”.

P001

Bc

Bonding-curve issuance

n

contested

P002

Tw

Time-weighted AMM execution

n

contested

P003

Da

Dutch-auction descent

n

contested

P004

Cg

Credit delegation

n

contested

P005

Ir

Insurance reserve fund

n

contested

P006

Zk

Verifiable state proof

n

contested

P007
Ve
Vote-escrow
allocation



P010
Sq
Shared ordering
commitment



P008
Kg
Credential-gated
transfer



P009
Ua
Unified-balance
ledger



HOW TO THINK WITH IT

Six questions, in this order. Run them against anything — a protocol you are auditing, a pitch deck, your own design — and the table stops being a diagram and becomes a method. Question three pays for the other five.

1

What are its parts?

Name every recurring financial state transition and refuse everything else. ERC-4626 is a socket, not a mechanism. A hook is somewhere to put an element, not an element. “Points” is marketing. Most whitepapers describe three real mechanisms and forty pages of scenery.

3

Where does it read the outside world?

This is the question that pays. Every catastrophic loss this table explains runs through the price, not the code. Ask what it would cost to move that input — measured against pool depth, never against the attacker’s balance — and if the answer is less than what can be borrowed against it, the protocol is already lost.

5

Who eats the loss?

For every fee, delay, slippage and advance, name the claim class or the underwriter. An unassigned loss has not vanished — somebody is carrying it without being paid to, and usually it is the depositor who never read this far. If the documentation cannot name them, that is the answer.

2

What did each part force?

Parts are not chosen freely — each one drags in the next. Leverage forces a truth source, a solvency test and a way for the position to end. A protocol with leverage and only two of those three is not innovating. It is missing a part, and you have just found it.

4

Does anything feed itself?

Reflexive collateral is the most reliable way to lose everything ever deployed on-chain. A token that is simultaneously the collateral, the market that prices it, and the capital backing it has three defences that all fail on the same news. This is the one shape that goes to zero rather than merely down.

6

What did the formula fail to say?

The residue, and it is the most useful line on the page. For a lending market it is small. For anything touching real-world assets it is larger than the formula and contains the actual risk. A decomposition that reports no residue was not finished.

LEGEND

Stratum S0 → S4



Prerequisite depth, not risk. Carried by the band, the S-token and lightness — never hue.

■ core

□ candidate: recurrence evidence short

contested: note required



/ degenerate limit: not an element



no denominator — failures counted, survivors never were

■ hazard: forbidden

■ hazard: elevated

■ hazard: unverifiable

Hazard membership is categorical. It is never a probability, because no exposure or survivor counts exist.

—i→ Interface — Can these exchange the required calls, tokens, messages, proofs?

—e→ Economic — Do combined incentives, liquidity and loss allocation stay solvent?

—t→ Trust — Who can attest, censor, upgrade, pause, mint, freeze?

—n→ Informational — Who observes the order before it settles?

This table explains roughly two thirds of DeFi incidents by count and a minority of them by dollars. That is not a defect in the table — it is the most useful finding on this page. Your largest risk is not your financial design. It is your keys, your signing interface and your build pipeline. Get the mechanism right and you have eliminated the failures people write papers about, not the ones that actually took the money.